



Analysis and optimization of bending vibration mechanical properties of underfill BGA solder joints

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ABSTRACT

This paper establishes a finite element analysis (FEA) model for underfilled ball grid array (BGA) solder joints made of SAC305 solder balls, and analyzes their stress-strain behavior under combined bending-vibration loads. An experimental platform was constructed to measure strain in the underfilled BGA package under bending-vibration loading conditions, thereby validating the simulation accuracy. The study employs Response Surface Methodology (RSM), Principal Component Analysis (PCA), and Particle Swarm Optimization (PSO) to optimize solder joint structural parameters, enhancing their mechanical performance. The results indicate the optimal parameter combination consists of: solder bump diameter of 0.47 mm, solder joint height of 0.37 mm, pad diameter of 0.34 mm, and ball pitch of 0.80 mm. Compared with the initial configuration, the optimized solder joint interconnection structure achieves reductions of 25.61% in maximum equivalent stress and 44.85% in maximum equivalent strain, while exhibiting increases of 0.14% in first-order natural frequency and 0.26% in second-order natural frequency relative to the pre-optimized state.

1. Introduction

With the development of electronic packaging technology, the ball grid array (BGA) package interconnection structure faces new challenges, such as high-density interconnections, fine pitch, and the demand for high-end I/O development [1]. BGA devices are connected to printed circuit boards (PCBs) via BGA solder joints, which play a crucial role in providing a stable bridge between electronic components and the PCB [2]. In practical operating environments, BGA solder joints may experience combined bending and vibration loads, which generate a complex stress-strain state within the joints, accelerate crack initiation and propagation, reduce fatigue life, and ultimately compromise the reliability of BGA solder joint connections. Studies have shown that in structural dynamic optimization and design analysis, it is critical to consider the first and second natural frequencies or modes, as they play an essential role in preventing damage caused by resonance [3]. To improve the reliability of BGA solder joints under complex conditions such as bending and vibration, methods such as bottom-filling reinforcement and optimization of solder joint structural parameters can be used to effectively reduce solder joint stress and strain, while enhancing the first and second natural frequencies [4].

Currently, some researchers have explored the reliability of solder joints under bending and vibration loads. For example, Kelly et al. [5] established a global strain analysis model through finite element analysis and Spearman correlation analysis, which can reflect the correlation between global bending strain and local strain at critical solder joints. Liu et al. [6] used finite element simulations and random vibration tests to demonstrate that the solder joint at the outermost corner of the BGA package exhibits higher peel stress than other joints. Liu et al. [7] also investigated the dynamic characteristics of PCBA under thermal-vibration combined loading through modal experiments and finite element simulation, and found that elevated temperature increases the peak displacement of PCBA, thereby shortening the fatigue life of solder joints. Wu et al. [8] established a stacked BGA solder joint model through finite element analysis and investigated its stress-strain distribution under coupled bending-vibration conditions, finding that the maximum stress is located at the outer edge of the lower surface where the lower solder ball contacts the PCB. Wei et al. [9] combined FEA, Taguchi orthogonal design, and the Harris-Hawk optimization algorithm to optimize QFN solder joint parameters, reducing bending-vibration coupled stress by 34.8%. Das et al. [10] improved hybrid SAC305-SnBi solder joints formed by low-temperature reflow

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through annealing, achieving longer service life under vibration or cyclic bending tests than unannealed SAC305 joints. The aforementioned studies indicate that traditional research on solder joint reliability has primarily focused on stress and strain indicators, without fully considering the influence of the first and second natural frequencies on the dynamic response and durability of solder joints.

For the evaluation of multiple mechanical indicators of solder joints, reducing the numerous indicators to a few key ones can lower analysis complexity. Principal component analysis is a dimensionality reduction method capable of achieving this goal and has been widely applied across various fields due to its powerful dimensionality reduction capability. For instance, Alkhayrat et al. [11] used PCA to reduce 220-dimensional telecom customer data to 20 dimensions, thereby reducing the computational complexity of subsequent K-means clustering. Hamidi et al. [12] reduced the dimensionality of DWT-decomposed signal coefficients using PCA and combined it with a fuzzy inference system, enabling simultaneous determination of multiple drug concentrations with high accuracy and strong predictive performance, serving as an alternative to HPLC. Zhang et al. [13] reduced 28 growth and physiological indicators to six principal components using PCA and membership function analysis, establishing a comprehensive evaluation model that enabled effective classification and screening of target germplasm. Meanwhile, it is noteworthy that no scholar has yet employed principal component analysis to perform dimensionality reduction on multiple key mechanical performance indicators of BGA solder joints in previous studies. To meet the optimization requirements of mechanical indicators in this study, the particle swarm optimization algorithm is adopted, which offers advantages such as simple implementation, fast convergence, and high precision [14–16].

To address the aforementioned research gap, this paper constructs a finite element model of an Underfill BGA solder joint and investigates the relationship between key structural parameters of the solder joint and four mechanical performance indicators (stress, strain, first and second natural frequencies) under combined bending and vibration loads. Furthermore, principal component-response surface analysis is employed to perform dimensionality reduction on these four mechanical performance metrics, converting them into normalized comprehensive scores to serve as comprehensive evaluation indicators. By introducing a particle swarm optimization algorithm to optimize the parameters of the quantitative model, optimal solder joint parameters are obtained,

thereby enhancing the mechanical performance of the solder joints. This provides a new perspective and analytical method for studying the mechanical performance of Underfill BGA solder joints under bending-vibration loads.

2. Bending-vibration finite element analysis

2.1. Finite element model establishment

This study selects the STM32F103 component with a BGA packaging structure, produced by STMicroelectronics, as the research subject. A 3D finite element model, as shown in Fig. 1, was established, consisting of five parts: PCB layer, BGA solder joints, BT substrate, chip (3×3), and bottom-filling layer. The nine chips in the model are of the same type. In the model shown in Fig. 1, the chip size is 10mm × 10mm × 0.65 mm; the BT substrate size is 10mm × 10mm × 0.1 mm; the solder ball grid is 12×12, with a solder ball height of 0.3 mm, pad diameter of 0.3 mm, solder ball diameter of 0.4 mm, and solder ball pitch of 0.8 mm, following the JESD22-B113B standard [17]. The PCB layer size is 132mm × 77mm × 1 mm, with the nine chips placed at the center of the PCB, and the chip pitch is 5 mm. The stress-strain relationship of the SAC305 material used in BGA solder joints is described by a linear elastic constitutive model. The material parameters for each part of the model are shown in Table 1.

During the meshing stage, SOLID185 elements were used for the solder balls in the model, while SOLID45 elements were used for the remaining parts. To improve computational efficiency and data accuracy, a mapping method was employed for meshing, and local mesh refinement was applied to the surfaces on both sides of the solder joints. Additionally, based on the first-order natural frequency of the finite element model as the criterion, five models with different mesh densities

Table 1
Material parameters.

Material	Elastic Modulus E/Gpa	Poisson's Ratio μ	Density $\rho/kg\cdot m^{-3}$
PCB	18.2	0.25	1880
SAC305	35.18	0.35	7390
Chip	15.5	0.28	1650
BT substrate	22.3	0.3	1200
Underfill layer	2.2	0.29	1120

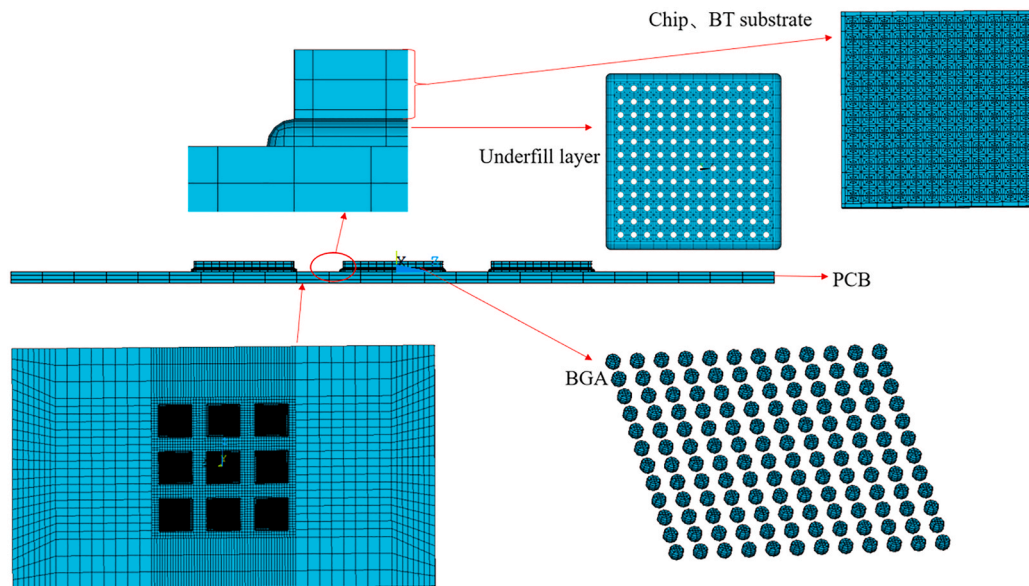


Fig. 1. 3D finite element model.

were created to investigate mesh independence [18]. The results are shown in Fig. 2. As the mesh becomes denser and the number of elements increases, the first-order natural frequency decreases and tends to stabilize. To improve computational efficiency without compromising data accuracy, the fourth mesh density scheme was adopted in this study. At this point, the entire model consists of 700,626 elements.

2.2. Stress and strain analysis of underfill BGA solder joints under bending-vibration load

This study uses an indirect coupling method to apply bending-vibration loads to the bottom-filled BGA solder joints. In the first step, as shown in Fig. 3, full displacement constraints are applied at the positions of the 1st line and 2nd line on the PCB (i.e., the symmetric support positions with a span of 105 mm). Then, a bending load of 1 mm in the positive Y-axis direction is applied at the position of the 3rd line (i.e., the center line) on the side of the PCB without chips. The results are then solved and saved. In the second step, the boundary conditions remain unchanged. The results from the first step are used as initial loads, and modal analysis is conducted to obtain the natural frequencies of the model. In the third step, based on the acceleration power spectral density curve from the U.S. military standard MIL-STD NAVMAT P9492 [19], a random vibration excitation in the Y-axis direction is applied to the model. A coupled solution analysis is then performed to obtain the stress-strain results of the BGA solder joints under bending-vibration composite loading. The acceleration power spectral density curve is shown in Fig. 4. When the random vibration frequency ranges from 20 Hz to 80 Hz, the curve has a positive slope of +3 dB/oct, and the corresponding acceleration power spectral density ranges from 0.01 to 0.04 g^2/Hz , with a value of 0.04 g^2/Hz at 80 Hz. When the random vibration frequency ranges from 80 Hz to 350 Hz, the corresponding acceleration power spectral density is 0.04 g^2/Hz . When the frequency is between 350 Hz and 2000 Hz, the curve decreases with a slope of -3 dB/oct, and the corresponding acceleration power spectral density ranges from 0.04 to 0.01 g^2/Hz .

Modal analysis and bending-vibration loading finite element analysis were performed on the simulation model. The first ten natural frequencies (Nat. Freq) of the solder joint interconnection structure are shown in Table 2. The first and second natural frequencies of the solder joint are 552.4 Hz and 996.3 Hz, respectively.

The stress distribution and strain distribution contour maps of the solder joint are shown in Figs. 5 and 6, respectively. As shown in Fig. 5, the maximum stress of the solder joint is 16.4 MPa, which is lower than the yield strength of SAC305 (43.9 MPa) measured by nanoindentation [20]. Meanwhile, the stress distribution in the solder joint is uneven, with the maximum equivalent stress occurring at the chip edge and the

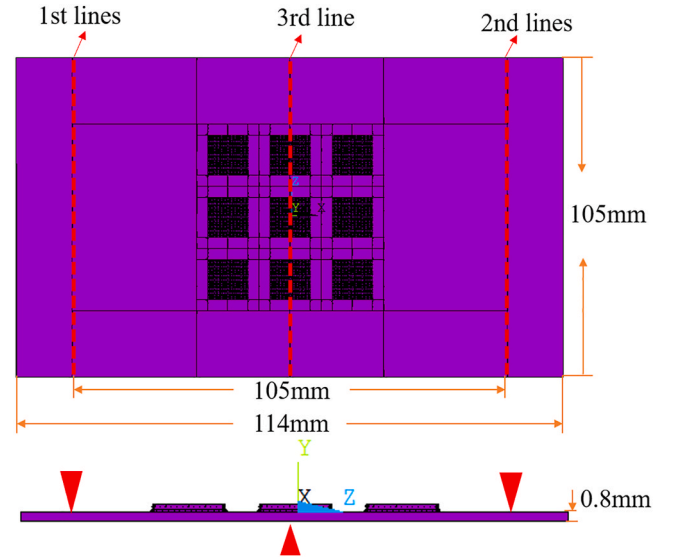


Fig. 3. Schematic diagram of bending-vibration load and constraints applied to the model.

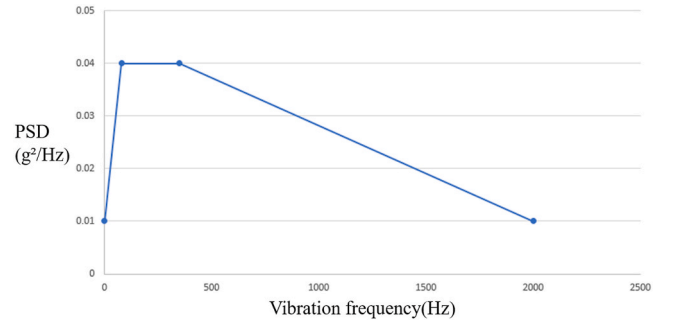


Fig. 4. Random vibration acceleration power spectral density (PSD) curve.

contact points between the solder joint and the PCB. As shown in Fig. 6, the maximum strain in the solder joint is 0.000466. The solder joint array exhibits significant deformation, with large strain regions on both sides of each chip. The magnitude of the deformation increases from the centerline outward.

3. Experimental design and validation

3.1. Design of bending-vibration load test

In this study, a bending-vibration load test platform was designed to test physical samples. Strain was measured and stress was calculated to compare with the results of finite element analysis, thereby validating that the simulation analysis can reliably simulate the experimental conditions.

The STM32F103 chip produced by STMicroelectronics was used to make the samples. First, the solder balls were removed from the STM32F103 samples. Then, the BGA rework station was used to re-ball (ensuring that the size parameters and material properties of the balls are consistent with the finite element analysis model). Next, the BGA rework station was used to design the reflow temperature curve and automatically solder the balls. The ball-attached chip was then mounted on the PCB board for the second soldering, as shown in Fig. 7. Finally, underfill was uniformly applied along the gap between the chip and the PCB board, and the sample was heated at 145 °C in an electric hot air drying oven for 15 min to cure.

Due to the small size of the BGA solder balls, strain gauges could not

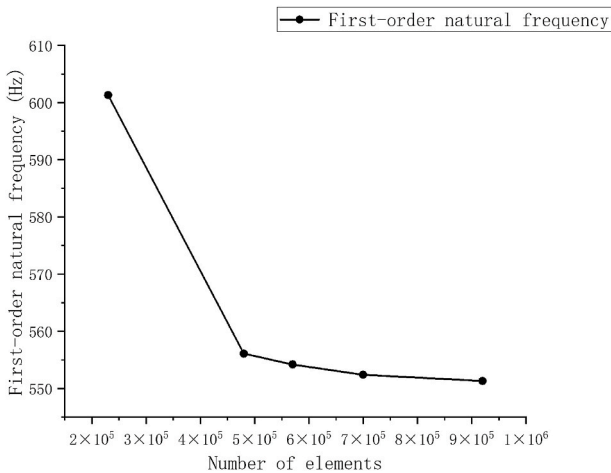
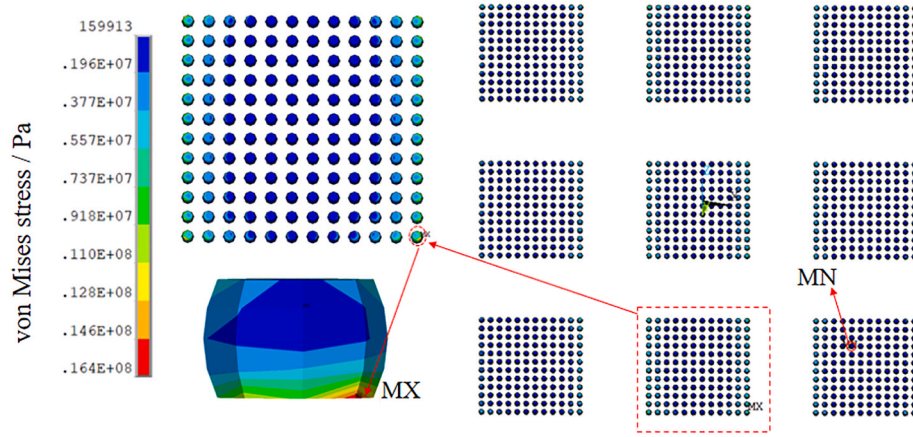
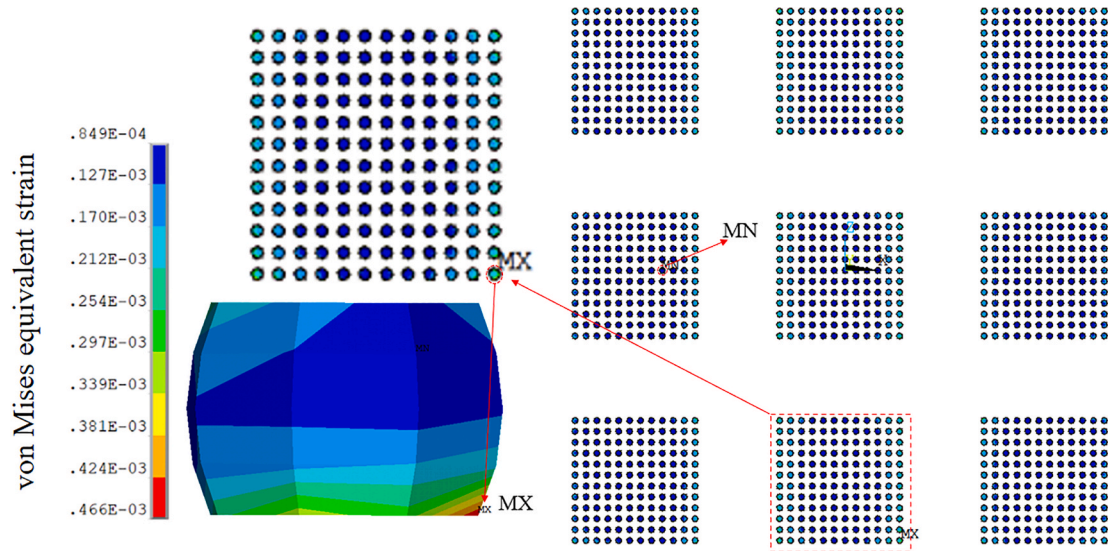


Fig. 2. Mesh independence verification.

Table 2

The top ten natural frequencies of the solder joint interconnection structure.

Mode	1	2	3	4	5	6	7	8	9	10
Nat. Freq (Hz)	552.4	996.3	1039.9	1309.9	1676.5	1793.1	1990.6	2162.7	2322.8	2490.6

**Fig. 5.** Solder joint stress distribution cloud diagram.**Fig. 6.** Solder joint strain distribution cloud diagram.

be directly attached for direct strain measurement. Therefore, the positions for attaching strain gauges in this study are shown in Fig. 8. Position 2 is selected at the center of the top surface of the chip package, and positions 1 and 3 are selected at the corners where the chip contacts the PCB board. A strain gauge of model BF120-1CA was used. Before bonding, the surface of the specimen was cleaned. Then, B-711 adhesive was evenly applied to the backside of the strain gauge, and the gauge was accurately positioned at the measurement location. Subsequently, a polytetrafluoroethylene (PTFE) film was placed over the strain gauge, and the excess adhesive was squeezed out by finger pressing. After curing at room temperature, the PTFE film was removed.

3.2. Comparison and analysis of experimental and simulation result

The bending-vibration coupling load test platform built in this study is shown in Fig. 9. The test platform consists of a bending-vibration load system and a strain measurement system. The test platform includes:

Vibration control software platform ①, dynamic strain signal testing and display platform ②, DH3817 dynamic resistance strain gauge ③, vibration controller ④, vibration control table ⑤, bending load application device ⑥, sample fixture ⑦, and vibration table ⑧.

The steps for measuring and processing strain data are as follows: First, the strain gauges at the test points of the sample are connected to the strain measurement system via signal cables. Then, according to the requirements of the simulation experiment, the sample was mounted on the vibration table ⑧ using the sample fixture ⑦ to complete the installation. Finally, a bending load was applied using device ⑥, which is a micrometer head with a pentagonal accessory fixed at the lower end of its measuring rod that maintains line contact with the PCB, and rotating the dial extends the measuring rod, driving the accessory to apply a 1 mm bending displacement to the PCB along the Y direction, while a vibration load excitation is applied through the vibration control software platform ①. During this process, the DH3817 dynamic resistance strain gauge ③ of the strain testing system recorded strain data at

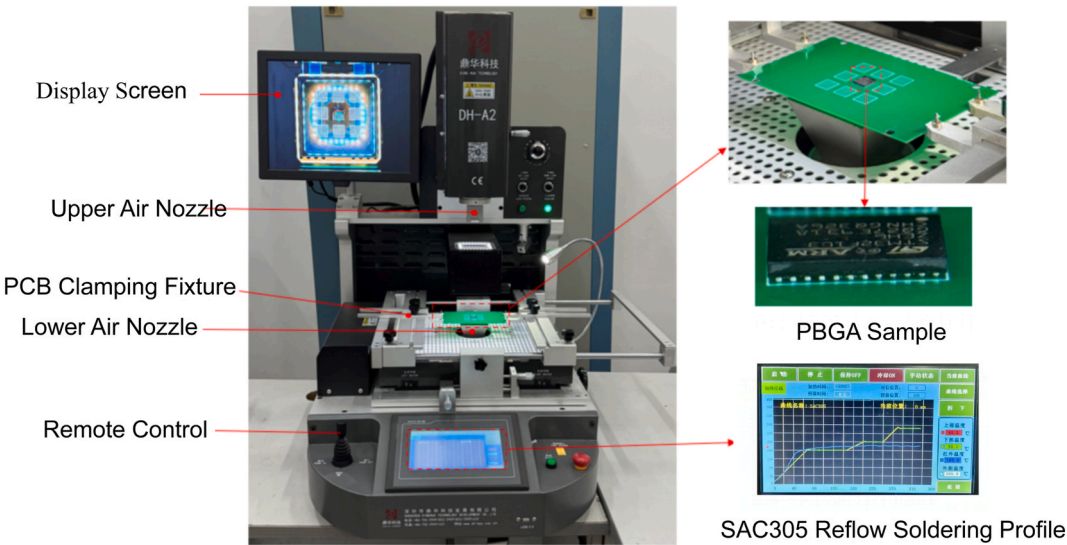


Fig. 7. Chip assembly.

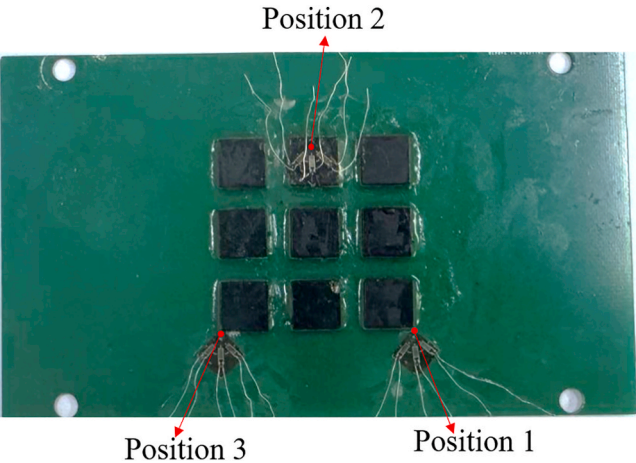


Fig. 8. Test point.

a sampling frequency of 5 kHz, obtaining dynamic strain signals in the

0°, 45°, and 90° directions at three measurement points. The stabilized strain values after the signals became steady were taken as the test results for each direction at each measurement point, and the strain data are presented in Table 3. The strain data were then substituted into equations (1)–(7) [21] to calculate the X-direction and Y-direction stress values and the shear stress in the XOY plane at each measurement point. The resulting stress data are presented in Table 4.

$$\varepsilon_{\theta} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} \cos 2\theta - \frac{\gamma_{xy}}{2} \sin 2\theta \tag{1}$$

$$\varepsilon_x = \varepsilon_{0^\circ} \tag{2}$$

Table 3
Linear strain measurement results at three test points.

Test point	Position 1	Position 2	Position 3
$\varepsilon_{0^\circ}/10^{-6}$	470.5	335.2	493.7
$\varepsilon_{90^\circ}/10^{-6}$	364.1	231.4	347.4
$\varepsilon_{45^\circ}/10^{-6}$	−224.8	−142.1	−228.1

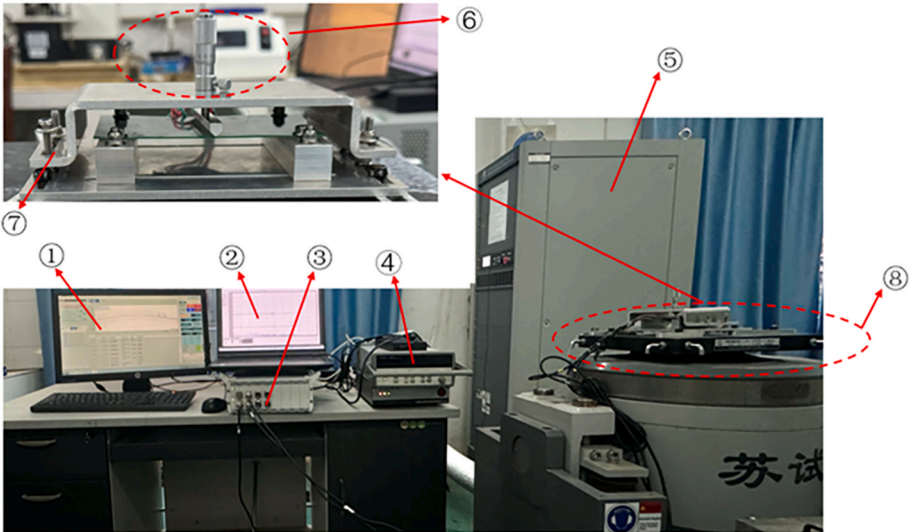


Fig. 9. Experimental platform.

Table 4
Comparison of equivalent stress values between tests and simulations.

Measurement Point	Test Point 1	Test Point 2	Test Point 3
X-direction Measured (Mpa)	10.90	7.63	11.27
X-direction Simulated (Mpa)	12.21	8.62	12.83
X-direction Error (%)	10.73	11.49	12.16
Y-direction Measured (Mpa)	9.35	6.12	9.14
Y-direction Simulated (Mpa)	10.19	5.85	10.26
Y-direction Error (%)	8.24	4.62	10.92
XOY Plane Measured (Mpa)	2.33	1.71	2.58
XOY Plane Simulated (Mpa)	2.87	1.86	3.12
XOY Plane Error (%)	18.81	8.06	17.31

$$\varepsilon_y = \varepsilon_{90^\circ} \quad (3)$$

$$\varepsilon_y = \varepsilon_y = \varepsilon_{90^\circ} + \varepsilon_{90^\circ} - 2\varepsilon_{45^\circ} \quad (4)$$

$$\sigma_x = \frac{E}{1 - \mu^2} (\varepsilon_x + \mu\varepsilon_y) \quad (5)$$

$$\sigma_y = \frac{E}{1 - \mu^2} (\varepsilon_y + \mu\varepsilon_x) \quad (6)$$

$$\tau_{xy} = \frac{E\gamma_{xy}}{2(1 + \mu)} \quad (7)$$

Where: $\varepsilon-\theta$ is the strain in any angular direction. Specifically, ε_{0° , ε_{90° , and ε_{45° are the strains in the plane 0° , 90° , and 45° directions respectively. ε_x , ε_y and γ_{xy} are the strains in the directions of X, Y and plane XOY respectively. σ_x and σ_y are the stresses in the directions of X and Y respectively. τ_{xy} is the shear stress in the plane of XOY, and γ_{xy} is the shear strain in the plane of XOY. E is the elastic modulus, and μ is the Poisson's ratio.

According to Table 4, the comparative analysis reveals that the relative errors between measured and simulated values at all three test points remain below 19%. Despite error sources such as installation deviations in strain gauge position/orientation and precision deviations in applying bending load via the micrometer head, the simulation results still demonstrate high accuracy.

4. Application of RSM and PCA to analyze the mechanical performance of solder joints under bending-vibration load

Under bending-vibration load conditions, the structural parameters of the solder joint affect the size and distribution of the stress and strain of the underfill BGA solder joint, and have a significant impact on the bending and vibration resistance performance of the solder joint. In this study, the solder joint height (H), solder joint diameter (D), pad diameter (R), and ball spacing (E) of the underfill BGA solder joint interconnection structure were selected as the research variables. Moreover, considering that the maximum stress, maximum strain, first-order natural frequency, and second-order natural frequency of the solder joint are very important under bending-vibration load [22], this study investigates the relationship between the above four solder joint parameters and these four performance indicators.

4.1. Response surface experimental design and results

Response Surface Methodology (RSM) is a commonly used statistical method for describing the complex relationship between design variables and target responses using polynomial functions [23]. This study employs the Central Composite Design (CCD) method in RSM, selecting four key solder joint structural parameters—solder joint height (H), BGA solder joint diameter (D), pad diameter (R), and solder ball pitch (E)—as key factors. Each factor has 5 levels, with the factor levels shown in Table 5 and the response surface combination table shown in Table 6.

Based on Table 6, a simulation model for each group of underfill BGA solder joints is established, and bending-vibration load analysis is carried out. The maximum stress, maximum strain, first natural frequency, and second natural frequency of the BGA solder joints in each model are obtained, with the simulation results presented in the last four columns of Table 6.

4.2. Multivariate regression analysis based on principal components

The four mechanical indicators of the solder joints in Table 6 reflect the mechanical performance of the solder joints under bending-vibration load conditions. Given the importance of these indicators, this study applies Principal Component Analysis (PCA) to reduce the dimensionality of these four mechanical performance indicators, forming a comprehensive mechanical performance evaluation index. The data from the Principal Component Analysis is shown in Table 7.

This study applies standard deviation normalization to the data in Table 7, converting Y1, Y2, Y3, and Y4 into dimensionless standardized values ZY1, ZY2, ZY3, and ZY4. The dimensionless standardized values are then subjected to KMO and Bartlett's test for sphericity. The KMO value is 0.644, and the Bartlett test significance result is less than 0.001, indicating that the indicators are correlated, suggesting that the data in Table 7 is suitable for Principal Component Analysis [24]. The eigenvalues, contribution rates, and cumulative contribution rates of the principal components obtained through Principal Component Analysis are shown in Table 8. As seen in Table 8, when the second principal component is extracted, the cumulative contribution rate is 89.037%, indicating that extracting the first and second principal components can represent the main information of the four indicators: maximum stress, maximum strain, first natural frequency, and second natural frequency of the bottom-filled BGA solder joints [25]. The eigenvector values of each indicator in the principal components are shown in Table 9.

Based on the loadings of each indicator on the principal components, the linear combination coefficients of each indicator on the different principal components are calculated, resulting in the score relationships for the first and second principal components, F_1 and F_2 , as shown in Equations (8) and (9).

$$F_1 = 0.233 \times ZY1 + 0.234 \times ZY2 + 0.203 \times ZY3 + 0.137 \times ZY4 \quad (8)$$

$$F_2 = -0.077 \times ZY1 + 0.077 \times ZY2 + 0.042 \times ZY3 + 0.200 \times ZY4 \quad (9)$$

Then, based on the variance contribution rate weights of the first and second principal components, and combining equations (8) and (9), the comprehensive score F value was calculated. To eliminate the differences in units and variation ranges of different indicators, the comprehensive score results were normalized to obtain the standardized normalized comprehensive score Z value. The specific results of the principal component comprehensive score and the normalized comprehensive score are shown in Table 10.

By combining the results of the normalized comprehensive scores from Tables 6 and 10, a response surface design analysis data table is obtained, as shown in Table 11.

Further, using the Design-Expert software with the normalized comprehensive score (Z value) as the response value—a comprehensive mechanical performance evaluation index, a nonlinear multivariate regression fitting analysis is performed on the solder joint diameter X_1 ,

Table 5
Factor level table.

Factors levels	Solder joint diameter D/mm	Solder joint height H/mm	Pad diameter R/mm	Ball spacing E/mm
1	0.4	0.30	0.26	0.74
2	0.42	0.32	0.28	0.76
3	0.44	0.34	0.30	0.78
4	0.46	0.36	0.32	0.80
5	0.48	0.38	0.34	0.82

Table 6

Response surface design and simulation results.

Num.	Solder joint diameter D/mm	Solder joint height H/mm	Pad diameter R/mm	Ball spacing E/mm	Maximum equivalent stress σ /MPa	Maximum equivalent strain $\varepsilon/(E-4)$	1st-order natural frequency f1/Hz	2nd-order natural frequency f2/Hz
1	0.44	0.34	0.26	0.78	14.6	4.16	551.15	994.68
2	0.44	0.34	0.3	0.78	15.3	4.34	551.14	994.81
3	0.42	0.32	0.32	0.76	15.8	4.5	551.4	991.56
4	0.46	0.32	0.28	0.76	15.2	4.39	551.62	993.72
5	0.44	0.34	0.3	0.74	14.9	4.24	550.55	989.18
6	0.44	0.34	0.3	0.78	15.3	4.34	551.14	994.81
7	0.44	0.34	0.3	0.78	15.3	4.34	551.14	994.81
8	0.42	0.32	0.32	0.8	16.3	4.62	551.91	997.07
9	0.44	0.34	0.3	0.82	15.6	4.45	551.61	1000.4
10	0.42	0.36	0.32	0.8	16	4.56	551.5	998.37
11	0.46	0.36	0.28	0.76	14.3	4.08	550.3	992.42
12	0.46	0.36	0.28	0.8	14.7	4.19	550.83	998.2
13	0.46	0.36	0.32	0.8	15.4	4.38	550.82	998.37
14	0.42	0.36	0.28	0.8	15.2	4.32	551.49	998.17
15	0.42	0.32	0.28	0.8	15.6	4.44	551.91	996.9
16	0.46	0.32	0.32	0.76	15.7	4.48	550.72	991.47
17	0.44	0.34	0.3	0.78	15.3	4.34	551.14	994.81
18	0.44	0.34	0.34	0.78	16	4.54	551.14	994.99
19	0.44	0.34	0.3	0.78	15.3	4.34	551.14	994.81
20	0.48	0.34	0.3	0.78	15.3	4.35	550.42	994.77
21	0.44	0.34	0.3	0.78	15.3	4.34	551.14	994.81
22	0.42	0.36	0.32	0.76	15.6	4.43	550.97	992.63
23	0.44	0.38	0.3	0.78	14.9	4.24	550.71	995.97
24	0.46	0.32	0.32	0.8	16.1	4.58	551.23	997.05
25	0.4	0.34	0.3	0.78	15.9	4.51	551.76	994.82
26	0.46	0.36	0.32	0.76	15	4.27	550.29	992.56
27	0.42	0.36	0.28	0.76	14.7	4.19	550.96	992.45
28	0.42	0.32	0.28	0.76	15.2	4.32	551.4	991.41
29	0.46	0.32	0.28	0.8	15.5	4.41	550.62	993.72
30	0.44	0.3	0.3	0.78	16.3	4.65	551.79	996.33

Table 7

Principal component analysis data.

Num.	Maximum equivalent stress (Y1) /MPa	Maximum equivalent strain (Y2) /(E-4)	1st-order natural frequency (Y3) /Hz	2nd-order natural frequency (Y4) /Hz
1	14.6	4.16	551.15	994.68
2	15.3	4.34	551.14	994.81
3	15.8	4.5	551.4	991.56
4	15.2	4.39	551.62	993.72
5	14.9	4.24	550.55	989.18
6	15.3	4.34	551.14	994.81
7	15.3	4.34	551.14	994.81
8	16.3	4.62	551.91	997.07
9	15.6	4.45	551.61	1000.4
10	16	4.56	551.5	998.37
11	14.3	4.08	550.3	992.42
12	14.7	4.19	550.83	998.2
13	15.4	4.38	550.82	998.37
14	15.2	4.32	551.49	998.17
15	15.6	4.44	551.91	996.9
16	15.7	4.48	550.72	991.47
17	15.3	4.34	551.14	994.81
18	16	4.54	551.14	994.99
19	15.3	4.34	551.14	994.81
20	15.3	4.35	550.42	994.77
21	15.3	4.34	551.14	994.81
22	15.6	4.43	550.97	992.63
23	14.9	4.24	550.71	995.97
24	16.1	4.58	551.23	997.05
25	15.9	4.51	551.76	994.82
26	15	4.27	550.29	992.56
27	14.7	4.19	550.96	992.45
28	15.2	4.32	551.4	991.41
29	15.5	4.41	550.62	993.72
30	16.3	4.65	551.79	996.33

Table 8

Eigenvalues and contribution rates of principal components.

Principal component	Eigenvalue	Contribution rate (%)	Cumulative contribution rate(%)
1	2.706	67.655	67.655
2	0.855	21.381	89.037
3	0.4333	10.819	99.855
4	0.006	0.145	100.00

Table 9

Eigenvector values of each indicator.

Target	1st principal component (eigenvector value)	2nd principal component (eigenvector value)
Maximum equivalent stress	0.345	-0.359
Maximum equivalent strain	0.346	-0.360
1st-order natural frequency	0.300	0.198
2nd-order natural frequency	0.203	0.934

solder joint height X2, pad diameter X3, and ball spacing X4 to obtain the multivariate regression model, which is shown in Equation (10).

Table 10

Principal component comprehensive score and standardized comprehensive score results.

Experiment number	1st principal component	2nd principal component	Comprehensive score F value	Normalized comprehensive score Z value
1	-0.73894	-0.01205	-0.56441	0.28787
2	-0.10406	-0.01146	-0.08183	0.50107
3	0.33995	-0.23249	0.20251	0.62669
4	0.08765	-0.00970	0.06428	0.56562
5	-1.02869	-0.50373	-0.90265	0.13844
6	-0.10406	-0.01146	-0.08183	0.50107
7	-0.10406	-0.01146	-0.08183	0.50107
8	1.30315	0.23848	1.04752	1.00000
9	0.73472	0.48751	0.67537	0.83559
10	0.94894	0.31615	0.79701	0.88933
11	-1.51585	-0.26709	-1.21602	0.00000
12	-0.59298	0.23697	-0.39371	0.36329
13	0.06089	0.24625	0.10539	0.58378
14	0.15335	0.29046	0.18627	0.61951
15	0.66166	0.23381	0.55893	0.78415
16	-0.04848	-0.29834	-0.10847	0.48930
17	-0.10406	-0.01146	-0.08183	0.50107
18	0.57163	0.00507	0.43560	0.72966
19	-0.10406	-0.01146	-0.08183	0.50107
20	-0.41006	-0.07596	-0.32984	0.39150
21	-0.10406	-0.01146	-0.08183	0.50107
22	-0.00541	-0.19582	-0.05113	0.51463
23	-0.58914	0.04666	-0.43648	0.34439
24	0.83781	0.18255	0.68048	0.83785
25	0.74099	0.04813	0.57463	0.79108
26	-0.86361	-0.26018	-0.71873	0.21970
27	-0.84698	-0.20445	-0.69271	0.23119
28	-0.25341	-0.25106	-0.25284	0.42552
29	-0.18287	-0.13795	-0.17209	0.46120
30	1.26005	0.18556	1.00206	0.97992

Table 11

Response surface design analysis data.

Number	X1 Solder joint diameter D/mm	X2 Solder joint height H/mm	X3 Pad diameter R/mm	X4 Ball spacing E/mm	Normalized comprehensive score Z value
1	0.44	0.34	0.26	0.78	0.28787
2	0.44	0.34	0.3	0.78	0.50107
3	0.42	0.32	0.32	0.76	0.62669
4	0.46	0.32	0.28	0.76	0.56562
5	0.44	0.34	0.3	0.74	0.13844
6	0.44	0.34	0.3	0.78	0.50107
7	0.44	0.34	0.3	0.78	0.50107
8	0.42	0.32	0.32	0.8	1.00000
9	0.44	0.34	0.3	0.82	0.83559
10	0.42	0.36	0.32	0.8	0.88933
11	0.46	0.36	0.28	0.76	0
12	0.46	0.36	0.28	0.8	0.36329
13	0.46	0.36	0.32	0.8	0.58378
14	0.42	0.36	0.28	0.8	0.61951
15	0.42	0.32	0.28	0.8	0.78415
16	0.46	0.32	0.32	0.76	0.48930
17	0.44	0.34	0.3	0.78	0.50107
18	0.44	0.34	0.34	0.78	0.72966
19	0.44	0.34	0.3	0.78	0.50107
20	0.48	0.34	0.3	0.78	0.39150
21	0.44	0.34	0.3	0.78	0.50107
22	0.42	0.36	0.32	0.76	0.51463
23	0.44	0.38	0.3	0.78	0.34439
24	0.46	0.32	0.32	0.8	0.83785
25	0.4	0.34	0.3	0.78	0.79108
26	0.46	0.36	0.32	0.76	0.21970
27	0.42	0.36	0.28	0.76	0.23119
28	0.42	0.32	0.28	0.76	0.42552
29	0.46	0.32	0.28	0.8	0.46120
30	0.44	0.3	0.3	0.78	0.97992

$$Z = 9930.25340 - 28801.42061X_1 - 27341.84072X_2$$

$$- 22533.84910X_3 - 11989.86909X_4 + 68380.31623X_1X_2$$

$$+ 53676.41939X_1X_3 + 35502.83559X_1X_4 + 76937.93121X_2X_3$$

$$+ 29077.35853X_2X_4 + 30288.48X_3X_4 + 10646.08137X_1^2$$

$$+ 5485.18337X_2^2 + 123.37X_3^2 + 1739975X_4^2 + 19028X_1X_2X_3$$

$$- 71555.63824X_1X_2X_4 - 75739.04070X_1X_3X_4$$

$$- 97670.04773X_2X_3X_4 - 3397.90220X_1^2X_2 + 6835.78644X_1^2X_3$$

$$- 14586.50549X_1^2X_4 - 13262.28619X_1X_2^2 + 242045X_1X_2X_3X_4$$

(10)

According to the analysis of variance results for the regression model in Table 12, the P-value of this nonlinear regression model is 0.0261, which is less than 0.05, indicating that the model is statistically significant. In addition, the lack-of-fit term for this model has a P-value of 0.9772 (>0.05), indicating that the lack of fit is not significant. The coefficient of determination (R^2) is 95.080%, demonstrating a good fit between the actual values and the predicted values of the regression model [26]. Therefore, this model is suitable for predicting the normalized comprehensive score. Based on the F-value and P-value results from the variance analysis of the nonlinear regression model, the ranking of the influence of the four solder joint parameters on the normalized composite score (Z) is as follows: $X_4 > X_1 > X_2 > X_3$, i.e., ball spacing > solder joint diameter > solder joint height > pad diameter.

In order to verify the credibility of the regression model, 15 different parameter combinations from Table 11 were substituted into the regression model equation (10) to obtain the predicted values Y of the normalized comprehensive scores for 15 solder joint models. The relative error (Rel. Err) data between the predicted values Y and the Z values of the verification group, as well as the 3D line chart, are shown in Table 13 and Fig. 10, respectively.

As seen in Table 13, the maximum absolute error between the predicted values Y and the simulation results Z for the 15 validation groups is 2.86%, indicating that the predicted values from the regression model are accurate and reliable, accurately reflecting the quantitative

Table 12

ANOVA results.

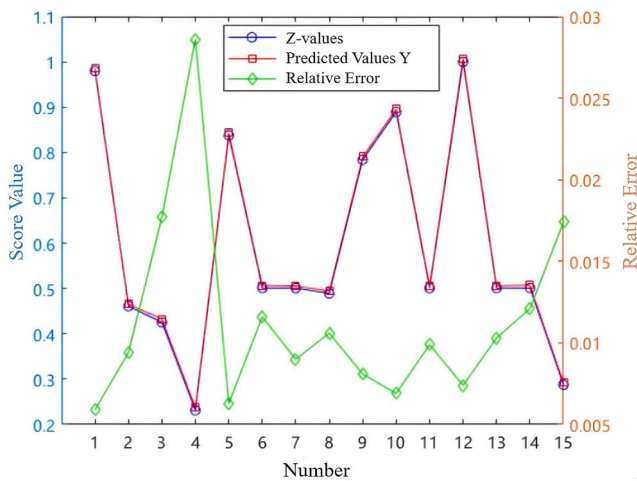
Source	Sum of Squares	df	Mean	F-value	P-value	significant
Model	1.6312	23	0.0709	5.0424	0.0261	*
X1	0.0233	1	0.0233	1.6563	0.2455	
X2	0.0096	1	0.0096	0.6813	0.4407	
X3	0.0095	1	0.0095	0.6749	0.4428	
X4	0.1076	1	0.1076	7.6474	0.0326	*
X1X2	0.0316	1	0.0316	2.2467	0.1846	
X1X3	0.0301	1	0.0301	2.1431	0.1935	
X1X4	0.2377	1	0.2377	16.8999	0.0063	**
X2X3	0.0252	1	0.0252	1.7927	0.2291	
X2X4	0.1497	1	0.1497	10.6468	0.0172	*
X3X4	0.0031	1	0.0031	0.2169	0.6579	
X1 ²	0.1181	1	0.1181	8.3980	0.0274	*
X2 ²	0.5383	1	0.5383	38.2716	0.0008	**
X3 ²	0.0668	1	0.0668	4.7497	0.0721	
X4 ²	0.0013	1	0.0013	0.0945	0.7690	
X1X2X3	0.0023	1	0.0023	0.1619	0.7014	
X1X2X4	0.0011	1	0.0011	0.0815	0.7849	
X1X3X4	0.0440	1	0.0440	3.1296	0.1273	
X2X3X4	0.0798	1	0.0798	5.6765	0.0546	
X1 ² X2	0.0039	1	0.0039	0.2802	0.6156	
X1 ² X3	0.0159	1	0.0159	1.1340	0.3279	
X1 ² X4	0.0726	1	0.0726	5.1635	0.0635	
X1X2 ²	0.0600	1	0.0600	4.2686	0.0843	
X1X2X3X4	0.0240	1	0.0240	1.7062	0.2393	
Lack of Fit	0.0000	1	0.0000	0.0009	0.9772	not significant

Note: * indicates a significant difference ($P < 0.05$); ** indicates a highly significant difference ($P < 0.01$).

Table 13

Comparison of predicted and simulated values.

Num.	X1 Solder joint diameter D/mm	X2 Solder joint height H/mm	X3 Pad diameter R/ mm	X4 Ball spacing E/ mm	Normalized comprehensive score Z value	Normalized comprehensive score predicted value Y	Rel. Err.
1	0.44	0.3	0.3	0.78	0.97992	0.98567	0.59%
2	0.46	0.32	0.28	0.8	0.46120	0.46555	0.94%
3	0.42	0.32	0.28	0.76	0.42552	0.43307	1.77%
4	0.42	0.36	0.28	0.76	0.23119	0.23780	2.86%
5	0.46	0.32	0.32	0.8	0.83785	0.84310	0.63%
6	0.44	0.34	0.3	0.78	0.50107	0.50688	1.16%
7	0.44	0.34	0.3	0.78	0.50107	0.50556	0.90%
8	0.46	0.32	0.32	0.76	0.48930	0.49451	1.06%
9	0.42	0.32	0.28	0.80	0.78415	0.79046	0.81%
10	0.42	0.36	0.32	0.8	0.88933	0.89545	0.69%
11	0.44	0.34	0.3	0.78	0.50107	0.50604	0.99%
12	0.42	0.32	0.32	0.80	1.00000	1.00735	0.74%
13	0.44	0.34	0.3	0.78	0.50107	0.50623	1.03%
14	0.44	0.34	0.3	0.78	0.50107	0.50714	1.21%
15	0.44	0.34	0.26	0.78	0.28787	0.29288	1.74%

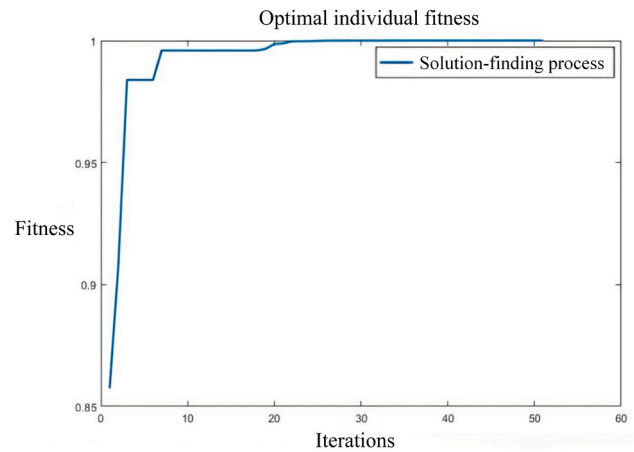
**Fig. 10.** Y value and Z value comparison line chart.

relationships between solder joint height, solder joint diameter, pad diameter, and solder ball pitch with the four mechanical indicators (solder joint stress, solder joint strain, first natural frequency, and second natural frequency).

4.3. Optimization analysis based on particle swarm algorithm

In order to reduce the maximum stress and strain of underfill ball grid array (BGA) solder joints and improve the structural natural frequencies (first and second order), particle swarm optimization is employed for parameter optimization. The fitness function is the regression model given by equation (10), i.e., $F(x) = Z$, with the goal of maximizing the normalized comprehensive score Z to optimize the key solder joint parameters. The constraints for the particle swarm algorithm are set as follows: $0.4 \leq X1 \leq 0.48$, $0.3 \leq X2 \leq 0.38$, $0.26 \leq X3 \leq 0.34$, $0.74 \leq X4 \leq 0.82$. The range of the normalized comprehensive score Z is limited to (0, 1). The initial population size is set to 100, the maximum number of iterations is 50, the maximum particle velocity is 0.1, and the learning factor is 2. The algorithm is considered to have converged if the change in the fitness value is less than 10^{-6} for 20 consecutive iterations. After 50 iterations, the iteration curve is plotted, as shown in Fig. 11.

As shown in Fig. 11, after 23 iterations, the fitness began to remain essentially constant, and the change in fitness was less than 10^{-6} for 20 consecutive iterations, satisfying the convergence criteria. At this point, a global optimal solution of approximately 0.99997 was obtained. Under

**Fig. 11.** Particle swarm algorithm iteration 50 times result diagram.

these conditions, the optimal parameter combination for the underfilled fine-pitch PBGA solder joint interconnection structure was determined as follows: solder bump diameter of 0.47 mm, solder joint height of 0.37 mm, pad diameter of 0.34 mm, and ball pitch of 0.80 mm. Based on the optimal parameter combination obtained through the particle swarm optimization (PSO) algorithm, a simulation model was established and subjected to combined bending-vibration loading for analysis. The resulting stress and strain contour plots of the solder joints are presented in Fig. 12(a) and (b), respectively. The simulation results demonstrate that after optimization, the maximum equivalent stress in the solder joint interconnection structure reached 12.2 MPa, with a maximum equivalent strain of $2.57\text{E-}4$. Furthermore, the first-order and second-order natural frequencies of the solder joint structure were measured at 553.15 Hz and 998.91 Hz, respectively.

Compared with the original unoptimized design, the optimized solder joint interconnection structure shows a 25.61% reduction in maximum equivalent stress and a 44.85% decrease in maximum strain, while maintaining favorable dynamic characteristics with slight increases of 0.14% and 0.26% in first-order and second-order natural frequencies (553.15 Hz and 998.91 Hz respectively). These comprehensive optimization results demonstrate improved mechanical performance, confirming that the parameter combination obtained through Response Surface Methodology combined with Principal Component Analysis and Particle Swarm Optimization effectively achieves multi-objective optimization for underfilled fine-pitch PBGA interconnection structures.

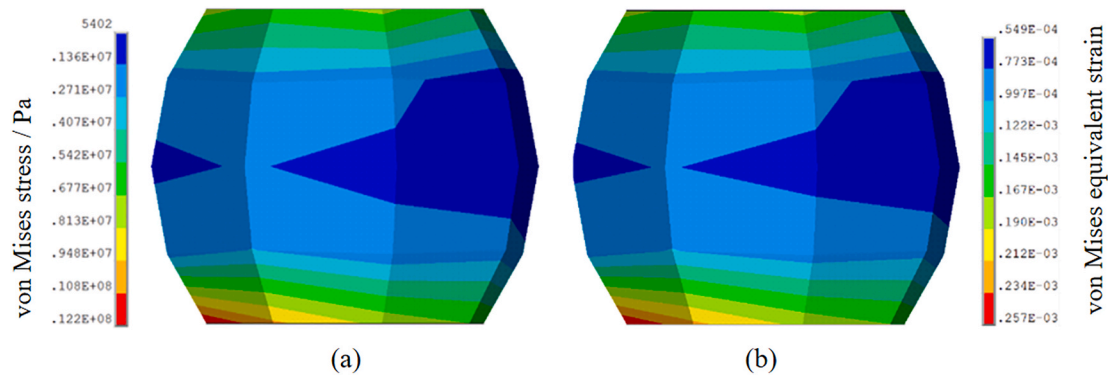


Fig. 12. Maximum stress and strain cloud diagram of the optimized solder joint.

5. Conclusions

- (1) The relative errors between the experimental and simulated stress values at the three test points are all less than 19%, indicating a high consistency between the experimental tests and the simulation analysis of the samples.
- (2) The order of influence of the four solder joint structural parameters on the comprehensive mechanical performance index, from greatest to least, is as follows: ball spacing > solder joint diameter > solder joint height > pad diameter.
- (3) At a 95% confidence level, the response surface design-principal component analysis regression model can accurately express the quantitative relationship between the solder joint height, solder joint diameter, pad diameter, and ball spacing and the four comprehensive mechanical performance indicators (solder joint stress, solder joint strain, first-order natural frequency, and second-order natural frequency).
- (4) Compared to the initial configuration, the optimized solder joint interconnection structure achieved significant reductions of 25.61% in maximum equivalent stress and 44.85% in maximum equivalent strain. Concurrently, the structure's first-order and second-order natural frequencies showed respective increases of 0.14% and 0.26% relative to the pre-optimization state.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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